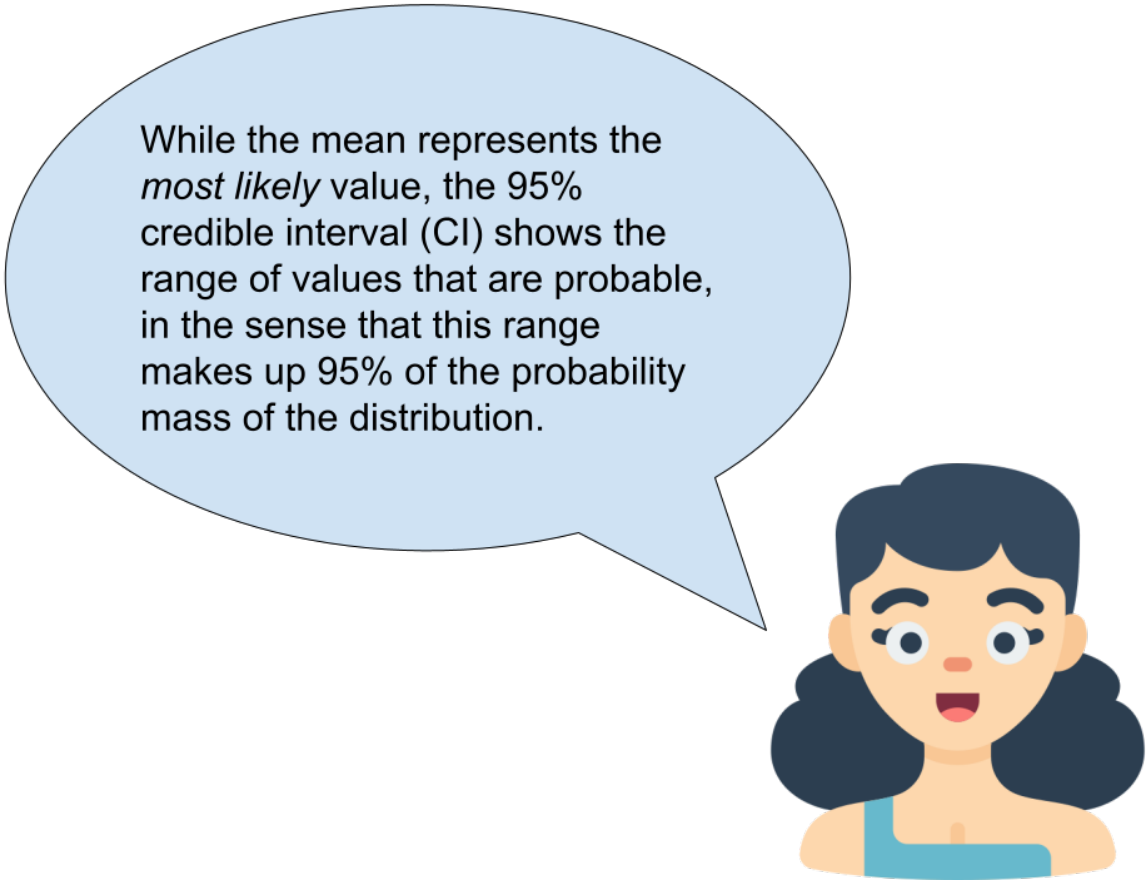
what's called the 95% credible interval, or CI (the shaded region).

Note that the acronym "CI" is often used to abbreviate "confidence interval" in frequentist statistics; throughout this book, we will use "CI" to denote "credible interval" exclusively. Many words can be said about the technical difference between the two concepts, but on a high level, we can still think of this CI as a range where it's likely that a quantity of interest (in this case the true value of the function we're predicting) falls into.

SIDEBAR **Gaussian process versus Ridge regression**

Interestingly, when using the same covariance function (also known as kernel), the mean predictions made by the GP matches exactly with those by ridge regression. (We will learn more about covariance functions in Chapter 3.) This is to say that a GP receives all the benefits that a ridge regression model possesses, while offering the additional CI predictions.



Effectively, this CI measures our level of uncertainty about the value at each test location. If a location has a large predictive CI (at -2 or 4 in Figure 1.4, for example), then there are a wider range of values that are probable for this value. In other words, we have higher uncertainty about this value. If a location has a narrow CI (0 or 2 in Figure 1.4), then we are more confident about the value at this location. A nice feature of the GP is that for each point in the training data, the

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